Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of 3D Perception and Transfer Learning-Based Robot Teaching Technology for Product Grasping and Contact-Type Manufacturing Tasks			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	*Robot Control and Intelligence Technologies*	Software	Software Solution
Concept of Proposed Technology	**1. Core Concept** We propose to develop novel methods to minimize training time of robot with a manipulator when the target object changes. To achieve this, three specific goals are to be achieved: - Development of 3D point cloud acquisition technology based on a 3DoF high-speed camera module - Development of 6D Pose Estimation Technology Based on 3D Point Clouds - Development of Transfer Learning Technology for Teaching Multi-Joint Robots Using Grippers with Various Grasping Shapes - Data Collection and Transfer Learning Technology Development for One Type of High-Difficulty Robot Motion **2. Key Features**			

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	2.1 3DoF High-Speed Camera Module-Based 3D Point Cloud Data Acquisition Technology - Development of a 3DoF high-speed camera module - Technology for acquiring high-density 3D point cloud data for industrial processed goods or products by size (ultra-small/small/medium) - Designing field of view/resolution/frame rate conditions optimized for robotic grasping, not just simple sensor use **2.2 3D Point Cloud-Based 6D Pose Estimation** - 3D Point Cloud-Based 6D Pose Estimation Technology for precise 6D Pose Estimation (position: x, y, z; orientation: roll, pitch, yaw) based on collected point clouds - Derivation of key performance metrics for pose estimation when size/shape/material vary **2.3 Transfer Learning for Teaching Product-Grasping Robots** - AI models learning latent skills/action embeddings - Reinforcement learning-based fine-tuning and teaching correction models **2.4 Research on multi-joint robot tasks related to adhesion/attachment** - Data modeling for fine control of contact pressure/velocity/angle - Analysis of surface material/curvature/cleanliness **3. Potential Use** - Smart / Autonomous Manufacturing - Logistics and Warehouse Automation - Service Robots and Collaborative Robots
Rationale for Joint R&D Project	**1. The Need for Joint R&D Projects** **1.1 Structural Limitations of Existing Robot Technology** Industrial and service robots largely depend on pre-programming, making adaptation to new environments or targets costly and time-consuming. As a result, robots function mainly as automation equipment and are sensitive to environmental changes. Advancing toward intelligent systems requires flexible technologies that can handle such changes for long-term operation. **1.2 Technological Demands Arising from Industrial and Societal Changes** Contemporary manufacturing, logistics, and service sectors are experiencing significant transitions: the broad implementation of high-variability, small-lot, multi-product production; an ongoing rise in the diversity and number of items requiring management; and increased challenges in recruiting personnel specialized in robotic operations. Consequently, there is a growing need for technologies that can flexibly adapt to these evolving requirements at a reasonable cost. **2. Why Collaboration with Leading International Research Institutes is Necessary** **2.1 Problem-Definition-Centric Research Drive Approach** This project goes beyond basic performance enhancements or the creation of applied technologies. It seeks to address fundamental questions, such as: “What is the optimal method for an articulated robot to acquire necessary data for object grasping?”; “How can we reduce instructional complexity when training a multi-joint robot in scenarios where the object or task evolves?”; and “For complex

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	actions like applying tape—beyond simple object manipulation—which data should be collected and what type of AI model is required?" The primary objective is to rigorously define these problems and systematically resolve them through data-driven approaches and artificial intelligence. **2.2 Comparative Advantages of Leading Overseas Research Institutions** Top overseas research institutions offer unique strengths due to their independent operations. Partnering with experienced organizations is essential for both expanding current research and developing new fields. **2.3 Limitations of Independent Execution and Synergy of International Collaboration** There are technical challenges when working independently, especially in forming core theories and clarifying problems. To address this, our company focuses on identifying practical industrial issues, collecting real-world data, building systems, and testing solutions in the field. Meanwhile, international research institutions concentrate on developing new theories, algorithms, and defining problems. By merging "field-validated theories" with "systems grounded in theory," we achieve strong synergy.
Key Words	1. 3D Point Cloud Data 2. 6D Pose Estimation 3. Transfer Learning for Teaching 4. Reinforcement Learning-based Fine-Tuning 5. latent skills/action embeddings

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	The proposed project will focus on creating a 3D sensing system, developing AI techniques for 6D pose estimation, advancing robot training with transfer learning, and using reinforcement learning to refine training methods. We also plan to improve technology for multi-joint robots working in environments with frequent contact. Together, these innovations are designed to help robots learn intricate manufacturing tasks directly from people and adapt those skills to various products and processes, ultimately boosting both flexibility and efficiency in automated manufacturing.			
R&D Target Specifications	**1. 6D Pose Estimation Position Error** - Definition: Average estimated error of object position (x, y, z) - Reference: x,y < 2 mm / z < 4 mm [1] - Target Value: ≤ 3 mm **2. Unseen Product Grasping Success Rate** - Definition: Grasping success rate for new products not included in training - Reference: 87.88% [2] - Target Value: ≥ 80% **3. Data Reduction Rate via Transfer Learning** - Definition: Reduction ratio of data required for teaching new product grasping - Reference: An average success rate of 70% for unseen object [3] - Target Value: ≥ 60% reduction **4. Failure Rate Reduction via RL-Based Teaching Calibration** - Definition: Comparison of grasping failure rates before and after applying RL calibration - Reference: one-shot imitation + residual learning success rate 46% enhanced, cycle time 25% reduced [4] - Target Value: ≥ 30% reduction **5. Real-time Inference and Control Latency (End-to-End Latency)** - Definition: Total processing time from 3D point cloud acquisition → 6D pose estimation → grasp/contact action determination - Reference: Pose estimation 109.5 ms [5] - Target value: ≤ 200 ms			
R&D Plan	**1. Research Approach** The project adopts a three-phase, stage-gated research approach, where each phase produces independently verifiable outputs while also serving as the technical foundation for the next phase. Here are key principles for the project. - Human-in-the-loop first: Start from human knowledge and experience - Generalization before autonomy: Ensure robustness before independence - Predictive and adaptive intelligence: Enable foresight and long-term evolution - Continuous validation: Regular testing in simulated and real environments **2. R&D Plan and Schedule** Total Duration: May 2026 ~ April 2029 (36 months) **2.1 First-Year Research Plan (May 2026 ~ April 2027):** 3DoF High-Speed Camera Module-Based 3D Point Cloud Data Acquisition Technology 1) Research Objectives - Establish a 3DoF high-speed camera module and data acquisition technology to obtain practically effective 3D data for robotic grasping - Build grasp-friendly 3D point cloud datasets according to product size			

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	(ultra-compact / compact / medium) 2) Detailed Development Topics and Content - Development of a 3DoF High-Speed Camera Module - High-Density 3D Point Cloud Data Acquisition Technology - Sensing Condition Design from a Robot Grasping Optimization Perspective **2.2 Year 2 Research Plan (May 2027 ~ April 2028):** 3D Point Cloud-Based 6D Pose Estimation & Transfer Learning for Teaching Product-Grasping Robots 1) Research Objectives - Acquire 3D point cloud-based high-precision 6D pose estimation technology - Establish a transfer learning-based AI model for product grasping teaching 2) Key Research Tasks and Methodologies - 3D Point Cloud-Based 6D Pose Estimation - Derivation of Pose Estimation Performance Metrics - Transfer Learning for Teaching Product-Grasping Robots **2.3 Third-Year Research Plan (May 2028 ~ April 2029):** Transfer Learning for Teaching Product-Grasping Robots & Multi-Joint Robot Tasks for Adhesion/Attachment 1) Research Objectives - Completion and generalization of grasp teaching transfer technology - Expanding research to post-grasping processes: contact-based and adhesive multi-joint robot tasks 2) Key Research Tasks and Methodologies - Advancement of Grasping Teaching Transfer Learning - Definition of Contact/Attachment Multi-Joint Tasks - Data Definition and Collection for Contact Control

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**1. Technical Contributions** - Design of core algorithms and models for robot perception, learning, and transfer - Development of theory and network architecture for 6D Pose Estimation based on 3D Point Clouds - Foundational research on Latent Skill/Action Embedding, Transfer Learning, and RL Calibration - Proposal of behavior representation modeling and learning frameworks for contact and attachment tasks - Theoretical validation of performance limitations and generalizability **2. Research Scope** - Interpretation of sensing data, design of perception and learning algorithms - Validation of research level for Teaching/Transfer/RL correction models - Experimental design, performance metric definition, statistical analysis - Conducting research meeting international academic standards **3. Resources to Be Provided** **3.1 Human Resources** - Professor (PI), Postdoctoral Researcher, PhD/Master's Degree Researcher **3.2 Research Infrastructure** - GPU Server, Simulation Environment, Research Robot Platform **3.3 Knowledge Assets** - Prior Research Results, Algorithm Baselines - International Research Network and Joint Paper Promotion - Contribution to academic dissemination and standardization efforts

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	4. Specific Tasks and Responsibilities - 3D point cloud data representation and quality analysis - Definition of grasp-friendly data characteristics - Development of 6D pose estimation models - Research on latent skill-based grasp teaching models - Transfer learning performance analysis - Advanced algorithms for grasp transfer learning - Behavioral modeling of contact/attachment tasks and RL calibration frameworks - Verification of the theoretical consistency of research results
Expected R&R of Korean SME Partner	**1. Technical Contributions** - Implemented recognition, learning, and transfer algorithms developed at the university as software modules applicable to real-world robot environments - Designed the mechanics, electrical systems, and communications for a 3DoF high-speed camera module and integrated it into the system - Implemented the actual control pipeline required for robotic grasping and contact tasks - Optimizing teaching/transfer/RL calibration structures to align with industrial robot control flows - Improving field suitability by incorporating noise, errors, and constraints arising in actual manufacturing environments **2. Development Tasks** **2.1 Hardware and Sensing System Development** - Fabrication of 3DoF high-speed camera module and integration into robot cell - Sensor placement and calibration optimized for robotic grasping - Integration of sensors for contact tasks (e.g., force/torque sensors) **2.2 Robot Software Development** - Implementation of 3D point cloud acquisition and preprocessing pipeline - Development of middleware linking 6D Pose Estimation results to robot control - Implementation of a real-time execution framework for grasp teaching and Transfer Learning models - Development of multi-joint robot control logic for contact/attachment tasks **2.3 System Integration and Stabilization** - End-to-End system integration of Perception–Decision–Action - Ensuring real-time processing performance, stability, and repeatability - Stabilization for long-term operation in industrial environments **3. Testing and Validation Activities** - The company is responsible for testing and validation based on actual manufacturing environment standards. - Testing of grasping success rate and repeatability by product size (ultra-small/small/medium) - Verification of the correlation between 6D Pose Estimation accuracy and grasping performance - Performance comparison experiments before and after Transfer Learning application - Verification of the effectiveness of RL-based Teaching correction (failure reduction rate, stability improvement) - Evaluation of quality metrics (attachment uniformity, failure rate) in contact/attachment tasks - Reliability verification through long-term repetitive testing in real environments **4. Resource Commitments** **4.1 Human Resources** - Robot software engineer - AI/ML engineer (responsible for model deployment and tuning) - System integration and control engineer - Testing, validation, and operational personnel **4.2 Equipment and Infrastructure** - Industrial or collaborative robot systems - Various types of grippers and tools for contact tasks

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	- 3DoF high-speed camera and peripheral sensors - Manufacturing process testbed and pilot line - Computing resources for development and testing **4.3 Data Resources** - Actual manufacturing product and process data - Gripping/attachment task logs and failure case data - Field condition information (material/surface/environment) **5. Collaboration Efforts** - Companies serve as the link between research, development, and demonstration through university partnerships - Regular technical meetings and joint design with university researchers - Providing field feedback on algorithm requirements - Participating in joint experiment design and performance metric definition - Joint patent applications and technology asset management - Supporting joint paper or technical report writing - Proposing industrial application scenarios and expansion directions for research outcomes

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**1. Technological Achievements** - Development of a 3DoF high-speed camera module and task-oriented 3D point cloud acquisition technology optimized for robotic manipulation - Achievement of millimeter-level 6D pose estimation accuracy robust to variations in product size, shape, and material - Establishment of transfer learning–based robot teaching technologies that enable reuse of grasping skills across different products - Integration of reinforcement learning–based teaching correction to improve robustness under real manufacturing conditions - Extension of robotic capability from grasping to contact-rich tasks such as adhesion and attachment using multi-joint robots **2. Commercialization Potential** - Reduction of robot teaching and reprogramming effort by 70% or more, lowering automation deployment costs - Creation of reusable robot perception and teaching software modules suitable for licensing or integration into existing robotic platforms - Enablement of robotic automation for high-mix, low-volume manufacturing, particularly benefiting small and medium-sized enterprises - Foundation for productized solutions in grasping, attachment, and contact-sensitive manufacturing processes **3. Industrial and Economic Impact** - Improved manufacturing productivity and operational flexibility through reduced manual intervention - Automation of labor-intensive and ergonomically challenging tasks, enhancing workplace safety - Transition of human roles toward supervisory and teaching-oriented functions - Strengthening of national competitiveness in robotics, AI, and smart manufacturing technologies **4. Value of Joint Academia–Industry Collaboration** - Accelerated transition from research outcomes to deployable systems - Generation of intellectual property, demonstrable prototypes, and validated industrial use cases - Advancement of technology readiness from research-level prototypes to field-tested solutions

6. References

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[1] Wang Pan, Feng Zhu, Yingming Hao, Limin Zhang, “Fast and precise 6D pose estimation of textureless objects using the point cloud and gray image”, Applied Optics, vol. 57, Sep, 2018

[2] Haiyuan Gui, Shanchen Pang, Xiao He, Teng Wang, Sibo Qiao, Luqi Wang, Shihang Yu, “High–performance grasp pose detection via point cloud serialization attention”, Pattern Recognition, vol. 171, Mar, 2026

[3] Puhao Li, Yingying Wu, Ziheng Xi, Wanlin Li, Yuzhe Huang, Zhiyuan Zhang, Yinghan Chen, Jianan Wang, Song-Chun Zhu, Tengyu Liu, Siyuan Huang, “ControlVLA: Few-shot Object-centric Adaptation for Pre-trained Vision-Language-Action Models”, Jun, 2025

[4] Chuang Wang, Chupeng Su, Sun Baozheng, Gang Chen, Longhan Xie, “Extended residual learning with one-shot imitation learning for robotic assembly in semi-structured environment”, Front. Neurorobot, vol. 18, April, 2024

[5] Snehal Dikhale, Karankumar Patel, Daksh Dhingra, Itoshi Naramura, Akinobu Hayashi, Soshi Iba, Nawid Jamali, “VisuoTactile 6D Pose Estimation of an In-Hand Object using Vision and Tactile Sensor Data”, Jan, 2026

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization